Module 3 Lecture - Claims & Validity

Research Methods in Psychology

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1 Overview

1.1 Lesson Overview

- Up until now, we have mostly focused on the more _____ and philosophical parts of why we conduct research
- Now, we are going to start to put more _____ names to the many moving parts of a study
- This is going to include talking about variables, claims, and validities - topics which are going to _____ over the remainder of the semester.

2 Variables

2.1 Definition

- Some _____ that can vary, with at least 2 levels/values
  - Contrast this with a _____, which does not vary at all
- Variables are important in both the _____ and the data stages of the theory-hypothesis-data cycle
- Variables (and their definitions within the context of the study) are core, critical parts of a _____ - and will play an important role in the validity of the results

2.2 Scales of Measurement

Likely a review from statistics course

- Every variable has some _____ of how it is measured:
- **Interval/Ratio/Continuous:** A variable has _____, known distances between each level/value, arranged in *known* order.
  - Example: Age, height in centimeters, point score on test.
- **Ordinal:** A variable has _____ distances between each level/value, arranged in a *known* order.
  - Example: Place in a foot race, Class rank

- **Categorical/Discrete:** A variable's levels/values are entirely _____ from one another, arbitrary (no) arrangement of order
  - Example: Experimental group vs. Control group

2.3 Measured vs. Manipulated Variables

- **Measured/Observed Variable:** a measurement of something that is not _____ or *directly* caused by the experimenter (but may be the result of something an experimenter did).
  - Often, due to ethical or _____ constraints, or simply because it is simply impossible to directly change
  - Example: SAT score, hair color, class enrollment
- **Manipulated/Experimental Variable:** a measurement of something that is directly _____ by an experimenter
  - Example: putting an individual in an intervention or control group.
- Some variables *can* be _____, depending on the setup of the study.

2.4 Constructs to Operational Variables

- **Construct/Conceptual/Latent Variables:** Abstract variables usually representing some _____ or notion that cannot be explicitly and directly measured
  - However, these usually *are* the concepts we actually care to draw _____ on.
  - Example: Depression, Academic achievement/aptitude
- **Operational/Measured Variables:** Concrete variables as measured through some _____ or sense
  - This is what we can directly _____ in the study
  - Example: PHQ-9, test score
- In most research, we _____ the construct to some direct measurement tool. However, the focus of our conclusions is *not* the measured variable, but rather, the _____ that it purports to measure.
  - Example: when I assign a test, do I care *only* about the plain number of items correct a student has? NO! I care about the content mastery that the items are representative of.
- Generally, when we talk about theories, we are trying to discuss _____; and when we are talking about study-specific hypotheses, we discuss _____ variables.

- Side note: The gap between operational to latent variables is an extremely important area that deals with _____ validity, which is different from the validity we will be discussing shortly. This area of study deals with Measurement and Test Theories, as well as psychometrics (my favorite).

3 The 3 Types of Claims

3.1 Definition

- A **claim**, simply, is any _____ for some idea, interaction, or relationship between two or more objects.
- Claims can be derived from scientific evidence, or any of the _____ of knowledge we discussed previously
  - But remember the pitfalls of some sources of information, and prioritize _____!
- Claims come in 3 flavors: frequency, _____, and causal
  - All three have varying _____ to be valid, which more strict requirements moving from frequency -> association -> causal
  - Good evidence of any claim is a _____ of the scale of measurement and the design of the study

3.2 Frequency Claims

- A **frequency claim** is one that deals with the rate, percent, or _____ of a certain characteristic or phenomenon.
  - Hint: a _____ sign is oftentimes a dead giveaway for this type of claim - but keep in mind that a "claim" is not necessarily made every time you see a percent.
  - Example: "roughly 25% of college students report feeling stressed about an upcoming assessment"
- These claims are usually only made upon some single observed and _____ variable or measurement, and often describes a yes-no proposition.
  - Example: 20 of 50 students like the dining hall food -> 20 students say yes to liking the food, 50 said no to the same question
- These claims are _____ dealing with a categorical scale of measurement
  - Stats sidebar: testing for claims of frequency are normally done with tests such χ^2 or Fisher's exact tests

- _____ represented by pie charts or bar charts

3.3 Association Claims

- An **association claim** is made regarding _____ or more variables, attempting to describe or quantify the relationship between them, *without* explaining which one might _____ the other
  - Hint: look for words like "correlate", "predict", "associated with", and/or "covary" to identify these claims
  - Example: I think that people who are more extroverted tend to be friendlier
- An association between two variables can either be described as *positive*, *negative*, or *non-existent/zero:*
  - _____ -> as one variable increases, so does the other
  - Negative -> as one variable increases, the other _____
  - Non-existent/zero -> both variables do not consistently _____
- Variables in this claim usually have to be at least _____, but are more commonly continuous in nature. In most cases, variables in this sort of study would only be _____ or measured.
  - Stats sidebar: testing for these claims is usually done with some form of correlation test (e.g., Pearson's r, Spearman's ρ, Kendall's $\tau - b$) or regression (e.g., linear - OLS, curvlinear)
- Graphically represented by _____ (one variable on x-axis, other on y-axis)

3.4 Causal Claims

- Similar to an association claim, with the added caveat that these claims make some _____ on which of two or more variables causes another.
  - This claim requires the most _____ and substantial evidence of all three claim types. Whereas a claim of association or frequency are largely *descriptive*, a causal claim is more *inferential*.
  - Example: I think eating more carbs makes me run better in the morning
- Similar to associative claims, a causal claim is likely to describe the movement of one variable in _____ to another, but must meet three additional criteria:
  - The two variables are correlated (valid _____ claim)
  - One variable clearly came before the other (_____ precedence)

  - The relationship or change cannot be _____ by some other factor/confound (internal validity - more on this later)
- For solid causal claims, _____ are really the most (and maybe only) acceptable design (but we will cover that later as well)
- In causal claims, we are most likely to encounter observed _____ data as our outcome, and some form of _____ experimental/-manipulated variable.
  - Stats sidebar: because of the scale of measurement just described, we are usually thinking about some sort of t-test or _____, to test means between tow or more groups.

3.5 Claims Are Not Always Research

- Claims can appear in personal experience, appeals to _____, and intuition; just because a claim is made, does not mean that they are based in research
  - *Example:* "I swear this happens to me every day", "This all happened because [insert reason here]"
  - What types of claims are those two above?
- If a claim is made, ask yourself, "what is the _____ of the claim?" Hopefully an empirical article!

4 Assessing Claims with 4 Types of Validity

4.1 Overview

- Alongside the theory-hypothesis-data framework we discussed early in the semester, and the reading strategy I offered for empirical articles - these 4 _____ will make up the backbone on how you approach critiquing scientific claims.

4.1.1 Construct Validity

- **Construct validity** is all about how well the variables under study are _____, manipulated, and measured. This is often closely related to the operational measures one chooses to capture the latent/construct variables.

- Faults in _____ validity imply a failure to properly represent the constructs under study, and will lead to a disconnect between the study and the _____ under study
- We will discuss more about "good" measurement _____ in week 5

4.1.2 External Validity

- **External validity**, also sometimes called _____, is all about how well the claims and results of a study/hypothesis apply to the broader _____ and other contexts. This tends to be primarily determined by how tightly controlled a study is (more control = less generalizable to daily life) and how individuals were _____ for a study (more on this in week 7).
- Faults in external validity lead to _____ in how well a claim can be extrapolated and abstracted outside the context of a study.
- Example: Is a tightly-controlled clinical study on mice in a lab cage readily applicable to the average person in their daily life? No!

4.1.3 Statistical Validity

This will be covered further in-depth in PSY-350

- **Statistical Validity** is all about whether the _____, p-values, effect sizes, and _____ methods support the hypothesis in question.
- Helpful questions to ask:
  - Did the authors state an α level in the methods?
  - Did the paper report adequate statistical information to assess significance? (e.g., p-values, _____ intervals, effect sizes - ideally all 3)
  - Did the authors themselves actually _____ this information correctly?
  - Was the analysis _____ enough to detect an effect? (e.g., big enough sample)
- Failures in this type of validity _____ the conclusions made in the discussion, which comes from the results.

4.1.4 Internal Validity

- **Internal validity** is most closely related to how well confounds are _____ for and noted in a study. Researchers may use a combination of _____ controls (which we will discuss in this class) and statistical controls (which are more relevant to PSY-350).
- Failure to properly address these will result in _____ on the actual causation/association mechanism between two or more variables.
- This tends be be one of the _____ validities to fully meet, and one that many people fail to account for in intuition and personal experience. This is where the famous phrase comes from: "Correlation does not equal _____" - because people fail to account for all the necessary 3 components for causal claims.

4.2 Claim Types and Validity

- In my opinion, the validities don't necessarily need a tailor-made application to each type of claim. The one exception is that internal validity is not really applicable to _____ claims, and only very little to association claims.

4.3 How to Prioritize Certain Validities

- As we have discussed before, no study can be so well-made that it satisfies all validities and answers all questions about a _____. Instead, it is our goal to balance _____ and our available resources to create a reasonably good study that provides evidence for a part of a broader question.
  - This is important to remember both as we create research (research _____) and as we read and assess existing research (research _____)!
- I propose 4 areas to consider when choosing how to prioritize:
  - Type of _____ to be made (frequency, association, causal) - see previous sections for discussions on those.
  - Practicality: time, resources, funding
  - Ethics: sampling method, interventions, _____ of populations
  - Impact: how important and _____ are our findings
- But, there is no one panacea or perfect solution to balance these all - we often do research understanding the natural _____ of our work.

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms